



# **GCE A LEVEL MARKING SCHEME**

**SUMMER 2019**

**A LEVEL  
CHEMISTRY - UNIT 3  
1410U30-1**

## **INTRODUCTION**

This marking scheme was used by WJEC for the 2019 examination. It was finalised after detailed discussion at examiners' conferences by all the examiners involved in the assessment. The conference was held shortly after the paper was taken so that reference could be made to the full range of candidates' responses, with photocopied scripts forming the basis of discussion. The aim of the conference was to ensure that the marking scheme was interpreted and applied in the same way by all examiners.

It is hoped that this information will be of assistance to centres but it is recognised at the same time that, without the benefit of participation in the examiners' conference, teachers may have different views on certain matters of detail or interpretation.

WJEC regrets that it cannot enter into any discussion or correspondence about this marking scheme.

## UNIT 3: PHYSICAL AND INORGANIC CHEMISTRY

### MARK SCHEME

#### GENERAL INSTRUCTIONS

##### Extended response questions

A level of response mark scheme is applied. The complete response should be read in order to establish the most appropriate band. Award the higher mark if there is a good match with content and communication criteria. Award the lower mark if either content or communication barely meets the criteria.

##### Marking rules

All work should be seen to have been marked.

Marking schemes will indicate when explicit working is deemed to be a necessary part of a correct answer.

Crossed out responses not replaced should be marked.

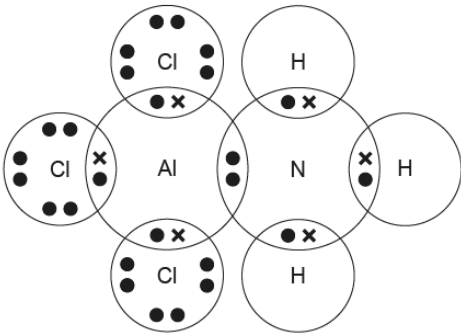
##### Marking abbreviations

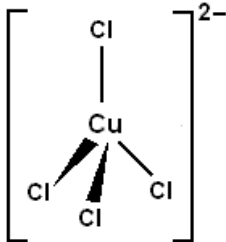
The following may be used in marking schemes or in the marking of scripts to indicate reasons for the marks awarded.

|     |   |                       |
|-----|---|-----------------------|
| cao | = | correct answer only   |
| ecf | = | error carried forward |
| bod | = | benefit of doubt      |

Credit should be awarded for correct and relevant alternative responses which are not recorded in the mark scheme.

## Section A

| Question |  |  |  | Marking details                                                                                                                                                                                             | Marks available |     |     |       |       |      |
|----------|--|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |  |  |  |                                                                                                                                                                                                             | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 1        |  |  |  | 3d <sup>7</sup><br>allow any clear representation                                                                                                                                                           |                 | 1   |     | 1     |       |      |
| 2        |  |  |  |  <p>NH<sub>3</sub> structure (1)</p> <p>dots and crosses showing coordinate bond and covalent bonds around Al atom (1)</p> |                 | 2   |     | 2     |       |      |
| 3        |  |  |  | $K_w = [H^+][OH^-]$                                                                                                                                                                                         | 1               |     |     | 1     |       |      |
| 4        |  |  |  | platinum as it is unreactive                                                                                                                                                                                | 1               |     |     | 1     |       |      |
| 5        |  |  |  | SiCl <sub>4</sub> has available d-orbitals whilst carbon does not (so water can bond to Si atom)                                                                                                            | 1               |     |     | 1     |       |      |
| 6        |  |  |  | forms a gas and gases have a greater entropy than liquids (and solids)                                                                                                                                      |                 | 1   |     | 1     |       |      |

| Question        |     |  |  | Marking details                                                                                                                                                                                                                | Marks available |     |     |       |       |      |
|-----------------|-----|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|                 |     |  |  |                                                                                                                                                                                                                                | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 7               | (a) |  |  | $\text{Fe}_3\text{O}_4 + 4\text{CO} \rightarrow 3\text{Fe} + 4\text{CO}_2$<br>ignore state symbols                                                                                                                             |                 | 1   |     | 1     |       |      |
|                 | (b) |  |  | award (1) for either of following <ul style="list-style-type: none"> <li>stability of +2 oxidation state increases down the group (due to the inert pair effect)</li> <li>carbon stable as +4 and lead stable as +2</li> </ul> | 1               |     |     | 1     |       |      |
| 8               |     |  |  |                                                                                                                                               |                 | 1   |     | 1     |       | 1    |
| Section A total |     |  |  |                                                                                                                                                                                                                                | 4               | 6   | 0   | 10    | 0     | 1    |

## Section B

| Question |     |       |  | Marking details                                                                                                                                                                                                                                                                                                                                                                 | Marks available |     |     |       |       |      |
|----------|-----|-------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                                                                                                                                                                                                                 | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 9        | (a) | (i)   |  | 121 kJ mol <sup>-1</sup>                                                                                                                                                                                                                                                                                                                                                        |                 | 1   |     | 1     |       |      |
|          |     | (ii)  |  | $\Delta H = -(-728) - (242) - (1735) - (178) + (-795)$ (1)<br>$\Delta H = -2222$ kJ mol <sup>-1</sup> (1)                                                                                                                                                                                                                                                                       |                 | 2   |     | 2     | 1     |      |
|          |     | (iii) |  | award (1) for any value in range 900-1600 kJ mol <sup>-1</sup><br><br>award (1) for either of following <ul style="list-style-type: none"> <li>ionisation energies increase for successive ionisation energies (so value must be more than half the ionisation of the first two electrons)</li> <li>second ionisation energy is greater than first ionisation energy</li> </ul> |                 | 1   | 1   | 2     |       |      |

| Question |     |       |  | Marking details                                                                                                                                                                                                                                                                                                                        | Marks available |     |     |       |       |            |
|----------|-----|-------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------------|
|          |     |       |  |                                                                                                                                                                                                                                                                                                                                        | AO1             | AO2 | AO3 | Total | Maths | Prac       |
|          | (b) | (i)   |  | at minimum temperature $\Delta G = 0$ so equation gives $\Delta S = \Delta H/T$ (1)<br>change units giving $T = 473 \text{ K}$ and $78000 \text{ J mol}^{-1}$ (1)<br>$\Delta S = 165 \text{ J K}^{-1} \text{ mol}^{-1}$ (allow 164.9) (1)<br>ECF possible                                                                              | 1               | 1   |     | 3     | 3     |            |
|          |     | (ii)  |  | 0.0284 mol of $\text{CaCl}_2$ (1)<br>0.1138 mol of water (1)<br>$x = 4$ (1)<br>ECF possible                                                                                                                                                                                                                                            |                 | 3   |     | 3     | 2     |            |
|          |     | (iii) |  | brick-red                                                                                                                                                                                                                                                                                                                              | 1               |     |     | 1     |       | 1          |
|          | (c) |       |  | $\text{CaCl}_2$<br>misty fumes (and white solid) (1)<br><br>$\text{CaBr}_2$<br>will produce orange fumes / solution as well as misty fumes (and white solid) (1)<br>accept will <b>also</b> produce orange fumes if first mark awarded<br><br>bromide is easier to oxidise than chloride (so can be oxidised by the sulfuric acid) (1) | 3               |     |     | 3     |       | 1<br><br>1 |
|          |     |       |  | Question 9 total                                                                                                                                                                                                                                                                                                                       | 5               | 9   | 1   | 15    | 6     | 3          |

| Question |     |       |  | Marking details                                                                                                                                                                                      | Marks available |     |            |       |       |      |
|----------|-----|-------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|------------|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                                      | AO1             | AO2 | AO3        | Total | Maths | Prac |
| 10       | (a) | (i)   |  | $\text{rate} = k[\text{CH}_3][\text{S}_2\text{O}_3^{2-}]$<br>award (1) for rate equation containing $k[\text{CH}_3]$<br>award (1) for second order rate equation                                     |                 | 1   | 1          | 2     |       |      |
|          |     | (ii)  |  | change $E_a$ to $81100 \text{ J mol}^{-1}$ (1)<br>rearrange equation to $T = -\frac{E_a}{\left(R \ln \frac{k}{A}\right)}$ (1)<br>$T = 346 \text{ K}$ (1)                                             | 1               |     | 1<br><br>1 | 3     | 3     |      |
|          | (b) | (i)   |  | quenching is the sudden stopping / significant slowing of a chemical reaction in a sample (1)<br>ensure sample composition does not change between taking sample and analysis OR during analysis (1) | 1<br><br>1      |     |            | 2     |       | 2    |
|          |     | (ii)  |  | starch                                                                                                                                                                                               | 1               |     |            | 1     |       | 1    |
|          |     | (iii) |  | canary/bright yellow                                                                                                                                                                                 | 1               |     |            | 1     |       | 1    |



| Question |     |      |    | Marking details                                                                                                                                                                                                                                                                                                                                                                     | Marks available |     |     |       |       |      |
|----------|-----|------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |    |                                                                                                                                                                                                                                                                                                                                                                                     | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          | (c) | (i)  | I  | catalyst in the same physical state as reaction                                                                                                                                                                                                                                                                                                                                     | 1               |     |     | 1     |       |      |
|          |     |      | II | award (1) each for any <b>two</b> of following <ul style="list-style-type: none"><li>transition metals have empty orbitals so can form bonds to reactant molecules / form complex</li><li>transition metals have variable oxidation states (so they can oxidise/reduce the reactants)</li><li>products are released and the metal returns to the original oxidation state</li></ul> | 2               |     |     | 2     |       |      |
|          |     | (ii) |    | $[H^+]^2 = K_a \times [\text{acid}]$<br>$[H^+] = 1.97 \times 10^{-3} \text{ mol dm}^{-3}$<br>pH = 2.7                                                                                                                                                                                                                                                                               |                 | 3   |     | 3     | 3     |      |
|          |     |      |    | Question 10 total                                                                                                                                                                                                                                                                                                                                                                   | 8               | 4   | 3   | 15    | 6     | 4    |

| Question |     |      |    | Marking details                                                                                                                                                                                                                                                                                                                                                                                              | Marks available |     |     |       |       |      |
|----------|-----|------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |    |                                                                                                                                                                                                                                                                                                                                                                                                              | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 11       | (a) | (i)  |    | $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O} \quad (1)$<br>$2\text{MnO}_4^- + 5\text{H}_2\text{C}_2\text{O}_4 + 6\text{H}^+ \rightarrow 2\text{Mn}^{2+} + 10\text{CO}_2 + 8\text{H}_2\text{O} \quad (1)$                                                                                                                                                     | 1               | 1   |     | 2     |       |      |
|          |     | (ii) | I  | 31.67 cm <sup>3</sup><br>accept 31.65 cm <sup>3</sup>                                                                                                                                                                                                                                                                                                                                                        |                 | 1   |     | 1     | 1     | 1    |
|          |     |      | II | moles of manganate(VII) = $\frac{31.67 \times 0.0505}{1000} = 1.599 \times 10^{-3} \text{ mol} \quad (1)$<br>ECF possible from mean volume<br>moles of oxalic acid = $1.599 \times 10^{-3} \times 2.5 = 4.00 \times 10^{-3} \text{ mol} \quad (1)$<br>ECF possible from equation<br>concentration of saturated solution = $\frac{4.00 \times 10^{-3}}{0.025} \times 10 = 1.60 \text{ mol dm}^{-3} \quad (1)$ |                 | 3   |     | 3     | 2     |      |

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Marks available |     |     |       |       |      |
|----------|-----|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          | (b) |  |  | <b>Indicative content</b><br>1. Oxalic acid has two acidic protons<br>2. The $K_a$ /acidity/ease of removal of each proton is different so they give two different vertical regions<br>3. First vertical region occurs at half the volume of second as each occurs after removing same number of protons<br>4. Flat regions from 5-10 and 20-25 due to formation of buffer<br>5. Alice, Brychan and David methods will work<br>6. Indicators only work if they change colour within a vertical region<br>7. Alice's method allows the first and second equivalence points to be found by plotting a curve<br>8. Brychan's indicator will change colour fully during the second vertical region<br>9. David's indicator will change colour fully during the first vertical region<br>10. Carys's method will not work as indicator will change colour gradually as sodium hydroxide is added |                 |     |     |       |       |      |
|          |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                 | 3   | 3   | 6     |       | 3    |

|  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |          |          |           |          |          |
|--|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|-----------|----------|----------|
|  |  |  | <p><b>5-6 marks</b><br/>The candidate includes at least six relevant points<br/><i>The candidate constructs a relevant, coherent and logically structured account including key elements of the indicative content. A sustained and substantiated line of reasoning is evident and scientific conventions and vocabulary is used accurately throughout.</i></p> <p><b>3-4 marks</b><br/>The candidate includes at least five relevant points<br/><i>The candidate constructs a coherent account including many of the key elements of the indicative content. Some reasoning is evident in the linking of key points and use of scientific conventions and vocabulary is generally sound.</i></p> <p><b>1-2 marks</b><br/>The candidate includes at least three relevant points<br/><i>The candidate attempts to link relevant points from the indicative content. Coherence is limited by omission and/or inclusion of irrelevant material. There is some evidence of appropriate use of scientific conventions and vocabulary.</i></p> <p><b>0 marks</b><br/><i>The candidate does not make any attempt or give an answer worthy of credit.</i></p> |          |          |          |           |          |          |
|  |  |  | <p style="text-align: right;"><b>Question 11 total</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>1</b> | <b>8</b> | <b>3</b> | <b>12</b> | <b>3</b> | <b>4</b> |

| Question |     |       |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks available |     |     |       |       |      |
|----------|-----|-------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                         | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 12       | (a) | (i)   |  | all four carbonates (1)<br><br>acids react with metal carbonates to form carbon dioxide gas (1)                                                                                                                                                                                                                                                                                                                                                         | 2               |     |     | 2     |       | 2    |
|          |     | (ii)  |  | (hydrochloric acid would form) <u>insoluble</u> compound with $\text{Pb}^{2+}$ (1)<br><br>should use nitric acid / ethanoic acid (1)                                                                                                                                                                                                                                                                                                                    |                 |     | 2   | 2     |       | 2    |
|          |     | (iii) |  | award (2) for <b>all</b> four correct<br>award (1) for <b>any</b> two correct<br><br>$\text{Mg}^{2+}(\text{aq})$ white (precipitate)<br>$\text{Fe}^{2+}(\text{aq})$ (dark) green (precipitate)<br>$\text{Cr}^{3+}(\text{aq})$ (grey) green (precipitate)<br>$\text{Pb}^{2+}(\text{aq})$ white (precipitate)                                                                                                                                             | 2               |     |     | 2     |       | 2    |
|          |     | (iv)  |  | add excess sodium hydroxide (1)<br><br>award (1) for either of following <ul style="list-style-type: none"> <li>magnesium hydroxide white precipitate remains but lead hydroxide precipitate dissolves (giving a colourless solution)</li> <li>iron(II) hydroxide green precipitate remains but chromium hydroxide precipitate dissolves (giving a dark green solution)</li> </ul><br>lead and chromium are amphoteric (iron and magnesium are not) (1) |                 |     | 3   | 3     |       | 3    |

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                           | Marks available |            |     |       |       |      |
|----------|-----|------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                           | AO1             | AO2        | AO3 | Total | Maths | Prac |
|          | (b) | (i)  |  | $\text{Mg}^{2+} \quad \frac{91}{24.3} = 3.7449$<br>$\text{Ca}^{2+} \quad \frac{50}{40.1} = 1.2469 \quad (1)$<br>ratio is 3Mg: 1Ca $(1)$<br>formula must contain $4 \times \text{CO}_3$ so it is $\text{Mg}_3\text{Ca}(\text{CO}_3)_4 \quad (1)$<br>accept alternative correct representations e.g.<br>$\text{CaCO}_3.3\text{MgCO}_3$<br>$(\text{MgCO}_3)_3.\text{CaCO}_3$ |                 | 1          | 2   | 3     | 2     |      |
|          |     | (ii) |  | concentration = $1.25 \times 10^{-6} \text{ mol dm}^{-3}$<br>(from calcium or magnesium concentrations) $(1)$<br>moles of solid used = $\frac{220 \times 10^{-6}}{353} = 0.6232 \times 10^{-6} \text{ mol} \quad (1)$<br>volume = $0.50 \text{ dm}^3 \quad (1)$<br>allow any value in the range 0.498-0.501 $\text{dm}^3$                                                 |                 | 1<br><br>1 | 1   | 3     | 2     |      |
|          |     |      |  | Question 12 total                                                                                                                                                                                                                                                                                                                                                         | 4               | 3          | 8   | 15    | 4     | 9    |

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marks available |     |                         |       |       |      |
|----------|-----|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-------------------------|-------|-------|------|
|          |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | AO1             | AO2 | AO3                     | Total | Maths | Prac |
| 13       | (a) |  |  | <p>Elfed's reaction produces <math>\text{Cr}^{2+}</math> (1)</p> <p>dichromate(VI) can be reduced to <math>\text{Cr}^{3+}</math> and further to <math>\text{Cr}^{2+}</math> as <math>E^\ominus</math> values for these two processes are more positive than that for <math>\text{Zn}^{2+}/\text{Zn}</math> / EMF values for reactions are positive (2.09 V and 0.34 V) (1)</p> <p>Fatima's reaction produces <math>\text{Cr}^{3+}</math> (1)</p> <p>oxygen oxidises <math>\text{Cr}^{2+}</math> back to <math>\text{Cr}^{3+}</math> as <math>E^\ominus</math> value is more positive than that for <math>\text{Cr}^{3+}/\text{Cr}^{2+}</math> / EMF value for reaction is positive (0.82 V) (1)</p> |                 |     | 1<br><br><br>1<br><br>1 | 4     |       |      |

| Question |     |       |  | Marking details                                                                                                                                                                                                                                                    | Marks available |          |             |           |          |          |
|----------|-----|-------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|-------------|-----------|----------|----------|
|          |     |       |  |                                                                                                                                                                                                                                                                    | AO1             | AO2      | AO3         | Total     | Maths    | Prac     |
|          | (b) | (i)   |  | $K_c = \frac{[CrO_4^{2-}]^2 [H^+]^2}{[Cr_2O_7^{2-}] [H_2O]}$                                                                                                                                                                                                       |                 | 1        |             | 1         |          |          |
|          |     | (ii)  |  | 1 dm <sup>3</sup> contains 1000g of water (1)<br><br>$\frac{1000}{18.02} = 55.5$ in 1 dm <sup>3</sup> (1)                                                                                                                                                          | 1               | 1        |             | 2         |          |          |
|          |     | (iii) |  | $[H^+]^2 = \frac{K_c \times [Cr_2O_7^{2-}] \times [H_2O]}{[CrO_4^{2-}]^2}$ (1)<br><br>$[H^+]^2 = 2.356 \times 10^{-12}$ or $[H^+] = 1.535 \times 10^{-6}$ mol dm <sup>-3</sup> (1)<br><br>pH = 5.81 (1)<br><br>student is incorrect as this solution is acidic (1) |                 | 1        | 1<br>1<br>1 | 4         | 3        |          |
|          |     | (iv)  |  | if $K_c$ increases the reaction is shifting to the right (1)<br><br>if a reaction shifts to right when heated it must be endothermic (1)                                                                                                                           |                 | 1        | 1           | 2         |          |          |
|          |     |       |  | <b>Question 13 total</b>                                                                                                                                                                                                                                           | <b>1</b>        | <b>4</b> | <b>8</b>    | <b>13</b> | <b>3</b> | <b>0</b> |



**UNIT 3: PHYSICAL AND INORGANIC CHEMISTRY****SUMMARY OF MARKS ALLOCATED TO ASSESSMENT OBJECTIVES**

| <b>Question</b>  | <b>AO1</b> | <b>AO2</b> | <b>AO3</b> | <b>Total</b> | <b>Maths</b> | <b>Prac</b> |
|------------------|------------|------------|------------|--------------|--------------|-------------|
| <b>Section A</b> | <b>4</b>   | <b>6</b>   | <b>0</b>   | <b>10</b>    | <b>0</b>     | <b>1</b>    |
| <b>9</b>         | <b>5</b>   | <b>9</b>   | <b>1</b>   | <b>15</b>    | <b>6</b>     | <b>3</b>    |
| <b>10</b>        | <b>8</b>   | <b>4</b>   | <b>3</b>   | <b>15</b>    | <b>6</b>     | <b>4</b>    |
| <b>11</b>        | <b>1</b>   | <b>8</b>   | <b>3</b>   | <b>12</b>    | <b>3</b>     | <b>4</b>    |
| <b>12</b>        | <b>4</b>   | <b>3</b>   | <b>8</b>   | <b>15</b>    | <b>4</b>     | <b>9</b>    |
| <b>13</b>        | <b>1</b>   | <b>4</b>   | <b>8</b>   | <b>13</b>    | <b>3</b>     | <b>0</b>    |
| <b>Totals</b>    | <b>23</b>  | <b>34</b>  | <b>23</b>  | <b>80</b>    | <b>22</b>    | <b>21</b>   |